

Phase 2 K-Audit Chain K85-K101 Status Summary

Keeper

2026-05-21 Thursday 12:15 EDT (actual via date)

Phase 2 K-Audit Chain K85-K101 Status Summary

Overview

17 Phase 2 K-audits filed Thursday morning (2026-05-21 EDT). Single-day chain extension represents peak K-audit cadence in BST history (Phase 1 K-audit chain K1-K84 accumulated over 7+ weeks; Phase 2 chain K85-K101 in ~5 hours).

Phase 2 transition observation per Cal #70: post 5-family architectural closure, K-audit growth shifts from architectural-category to INSTANCE-level chapter-category.

Chain table (K85-K101)

K	Topic	Filed (EDT)	F1-F4	B1-B4	Phase 2 cluster
K85	T (Time Reversal) substrate-operation	Thursday morning	4.0/4	3.8/4	CPT-cluster
K86	C (Charge Conjugation) substrate-operation	Thursday morning	4.0/4	3.8/4	CPT-cluster
K87	CPT theorem (composition)	Thursday morning	4.0/4	3.9/4	CPT-cluster (BST TIER-1 FALSIFIER SET)
K88	$m_p/m_e = 6\pi^5 = C_2 \cdot \pi^{\{n_C\}}$	Thursday morning	3.9/4	3.8/4	SM-foundation
K89	CKM Jarlskog 0.3% match	Thursday morning	3.7/4	2.9/4	SM-foundation (T1444 vacuum-subtraction dependency)
K90	Five Absences cluster	Thursday morning	4.0/4	3.8/4	BST TIER-1 FALSIFIER SET (NEW pattern)
K91	Experimental Program (SP-30 + SP-29)	Thursday morning	4.0/4	3.8/4	BST TIER-1 FALSIFIER SET
K92	a_e (electron g-2) ppt precision	Thursday morning	4.0/4	3.8/4	CROWN JEWEL

K	Topic	Filed (EDT)	F1-F4	B1-B4	Phase 2 cluster
K93	SM Gauge Group SU(3)×SU(2)×U(1)	Thursday morning	3.9/4	3.8/4	SM-foundation
K94	Three Generations	Thursday morning	3.9/4	3.8/4	SM-content cluster
K95	Color/Quarks (SM-content)	Thursday morning	3.9/4	3.8/4	SM-content cluster
K96	Lepton Sector (SM-content)	Thursday morning	3.9/4	3.8/4	SM-content cluster
K97	D_IV ⁵ APG Ratification (foundational meta-audit)	Thursday 11:37→12:05 EDT	3.9/4	3.9/4	Foundational meta-audit
K98	Strong-Uniqueness Chapter Audit (Vol 0 Ch 9)	Thursday 12:05 EDT	3.95/4	3.83/4	Foundational chapter cluster
K99	Conservation Laws Chapter Audit (Vol 0 Ch 8)	Thursday 12:06 EDT	4.0/4	3.80/4	Foundational chapter cluster
K100	Methodology Stack Chapter Audit (Vol 0 Ch 10)	Thursday 12:08 EDT	3.95/4	4.0/4	Foundational chapter cluster
K101	Five Integers + N_max Chapter Audit (Vol 0 Ch 2)	Thursday 12:12 EDT	4.0/4	3.93/4	Foundational chapter cluster

Cluster organization

CPT-cluster (K85+K86+K87)

First Phase 2 cluster filed Thursday morning. T (T2433) + C (T2434) substrate-operation theorems + CPT composition. Captures BST TIER-1 FALSIFIER SET first entry.

SM-foundation track (K88+K89+K93)

Mass-ratio ($6\pi^5$), CKM matrix Jarlskog, SM gauge group derivations. Honest correction Thursday 10:15 EDT on K89 J_CKM dependency on T1444 vacuum-subtraction mechanism (Elie within-session Cal Mode 1 self-correction).

SM-content cluster (K94+K95+K96)

Three generations + color/quarks + lepton sector. Cluster pattern (NEW Phase 2 entry).

BST TIER-1 FALSIFIER SET (K87+K90+K91)

Three cluster K-audits constituting BST's primary falsification framework. K90 cluster pattern is NEW (predictions-cluster) vs. K87 individual-prediction pattern.

CROWN JEWEL (K92)

a_e ppt precision against CODATA — BST's tightest single prediction.

Foundational meta-audit cluster (K97-K101)

K97 + K98 + K99 + K100 + K101 = 5-chapter foundational cluster: - K97 = APG ratification (foundational meta-audit) - K98 = Strong-Uniqueness Theorem chapter (Vol 0 Ch 9) - K99 = Conservation Laws chapter (Vol 0 Ch 8) - K100 = Methodology Stack chapter (Vol 0 Ch 10) - K101 = Five Integers + N_max chapter (Vol 0 Ch 2)

Average F1-F4 across foundational cluster: 3.97/4 (peak audit-tier average). Average B1-B4 across foundational cluster: 3.89/4.

Status tier distribution

- PRE-STAGE (Cal grade-pass pending): K85, K86, K87, K88, K89, K90, K91, K92, K93, K94, K95, K96, K97, K98, K99, K100, K101 (17/17)
- AUDIT-PARTIAL-READY: 0 (Cal grade-pass cadence pending Thursday afternoon-Friday)
- RATIFIED: 0

Effective Phase 2 chain depth: 17 pre-stages filed; 0 ratified. Cal cadence over coming days advances pre-stages → audit-partial-ready → RATIFIED.

K-audit chain extension trajectory

Phase 2 chain length (Thursday EOD projected): K85-K101 (17 audits) Phase 2 chain coverage: Vol 0 chapters (2, 8, 9, 10) + SM (3+1 generations + color + leptons + gauge group) + foundational (CPT + falsifiers + crown jewel + APG ratification)

Estimated Phase 2 chain endpoint: K150-K200 across multi-month Year 1 chapter-driven cadence per Cal #70.

Remaining Vol 0 chapters needing K-audit pre-stage: Ch 3 (Substrate Operating System), Ch 4 (Isotropy Group), Ch 5 (Boundary Conditions), Ch 6 (Integer Web Principle), Ch 7 (The Operator Zoo) — K102-K106 candidates.

Cal review queue impact

Cal review queue (per Consolidated Brief filed Thursday 12:14 EDT): - 5 K-audit endorsements (K97 + K98 + K99 + K100 + K101) in Tier 2 - 3 RIGOROUSLY CLOSED verifications (T2440 + T2441 + T2442) in Tier 1 - Estimated Cal turnaround: rest of Thursday afternoon-Friday rolling

Cross-references

- Strong-Uniqueness Theorem v0.9.1: 4 RIGOROUSLY CLOSED + 9 RATIFIED + 1 ADVANCING (closest to v1.0 in BST history)
- K97 path to RATIFIED: weeks-to-few-months per Cal #80 endorsement
- Year 1 Launch Trio Status Integration v0.1: multi-lane v0.5 → v1.0 path
- Lyra Sessions 6+ Planning v0.1: weekend cadence for additional RIGOROUSLY CLOSED criteria

Cadence health metrics

- 17 K-audits in single morning = peak chain extension cadence
- 5 chapter-audits in 12 minutes (12:05-12:12 EDT) = peak Phase 2 cadence
- 62 commits today (Thursday) = ~11 commits/hour sustained ~5.5 hours
- Methodology stack at 15 layers (Cal v0.3 count) = peak architectural maturation
- Cycle-time peak ~2 minutes (Cal #79 → Vol 0 Ch 9 v0.3 correction) = peak multi-CI consensus speed

Status

Phase 2 K-Audit Chain K85-K101 Status Summary filed Thursday 12:15 EDT.

17 Phase 2 K-audits Thursday morning. 5-chapter foundational cluster (K97-K101) ready for Cal grade-pass cadence.

— Keeper, 2026-05-21 Thursday 12:15 EDT (actual via date)

Thursday 12:34 EDT — Extension to K102-K106 + Vol 0 K-audit COMPLETE

K102-K106 added (Vol 0 coverage completion)

Filed Thursday 12:19-12:21 EDT (after this Summary's initial filing at 12:15 EDT):

K	Topic	F1-F4	B1-B4
K102	Substrate Operating System (Vol 0 Ch 3)	3.88/4	3.78/4
K103	Isotropy Group (Vol 0 Ch 4)	3.95/4	3.95/4
K104	Boundary Conditions (Vol 0 Ch 5)	3.88/4	3.70/4
K105	Integer Web Principle (Vol 0 Ch 6)	3.80/4	3.75/4
K106	The Operator Zoo (Vol 0 Ch 7)	3.95/4	3.85/4

Vol 0 K-audit COMPLETE (10/10)

K97 + K98 + K99 + K100 + K101 + K102 + K103 + K104 + K105 + K106 = 10 K-audits for 10 Vol 0 chapters.

Average F1-F4 across Vol 0 cluster: 3.93/4 STRONG Average B1-B4 across Vol 0 cluster: 3.85/4 STRONG

Updated Phase 2 chain status (Thursday 12:34 EDT)

- 22 Phase 2 K-audits filed Thursday morning (K85-K106)
- All 22 at PRE-STAGE pending Cal grade-pass cadence
- Vol 0 K-audit coverage COMPLETE
- K97 RATIFIED path REVISED: ~few weeks per Cal #81

Cal Review Queue (Thursday 12:34 EDT)

Per Cal Review Queue Consolidated Brief updated Thursday 12:28 EDT: - Tier 1: 3 RIGOROUSLY CLOSED verifications (T2440 + T2441 + T2442) — Cal #81 preliminary AGREE - Tier 2: 10 K-audit endorsements (K97-K106) - Tier 3: Year 1 trio chapter Cal cadence - Tier 4: Cross-lane verification (Sessions 6-9 specs + planning docs)

Path forward (Thursday 12:34 EDT)

1. Cal grade-pass cadence Friday-next week rolling K97-K106
2. Lyra Sessions 6-9 weekend (Friday-Sunday) → Strong-Uniqueness v0.9.5
3. K-audit chain extension as Vol 1 + Vol 2 chapters mature (multi-week)
4. K97 RATIFIED ~few weeks → Strong-Uniqueness v1.0 endpoint
5. Venue submission ~2026-09 substantially achievable

— Keeper update, 2026-05-21 Thursday 12:34 EDT (actual via date; Phase 2 chain extension K102-K106 + Vol 0 K-audit COMPLETE)